

Lebesgue's Theorem *If the function f is monotone on the open interval (a, b) , then it is differentiable almost everywhere on (a, b) .*

Proof Assume f is increasing. Furthermore, assume (a, b) is bounded. Otherwise, express (a, b) as the union of an ascending sequence of open, bounded intervals and use the continuity of Lebesgue measure. The set of points x in (a, b) at which $\overline{D}f(x) > \underline{D}f(x)$ is the union of the sets

$$E_{\alpha, \beta} = \{x \in (a, b) \mid \overline{D}f(x) > \alpha > \beta > \underline{D}f(x)\}$$

where α and β are rational numbers. Hence, since this is a countable collection, by the countable subadditivity of outer measure, it suffices to prove that each $E_{\alpha, \beta}$ has outer measure zero. Fix rationals α, β with $\alpha > \beta$ and set $E = E_{\alpha, \beta}$. Let $\epsilon > 0$. Choose an open set $\mathcal{O}$ for which

$$E \subseteq \mathcal{O} \subseteq (a, b) \text{ and } m(\mathcal{O}) < m^*(E) + \epsilon. \quad (10)$$

Let $\mathcal{F}$ be the collection of closed, bounded intervals $[c, d]$ contained in $\mathcal{O}$ for which $f(d) - f(c) < \beta(d - c)$. Since $\underline{D}f < \beta$ on E , $\mathcal{F}$ is a Vitali covering of E . The Vitali Covering Lemma tells us that there is a finite disjoint subcollection $\{[c_k, d_k]\}_{k=1}^n$ of $\mathcal{F}$ for which

$$m^*\left[E \setminus \bigcup_{k=1}^n [c_k, d_k]\right] < \epsilon. \quad (11)$$

By the choice of the intervals $[c_k, d_k]$, the inclusion of the union of the disjoint collection intervals $\{[c_k, d_k]\}_{k=1}^n$ in $\mathcal{O}$ and (10),

$$\sum_{k=1}^n [f(d_k) - f(c_k)] < \beta \left[\sum_{k=1}^n (d_k - c_k) \right] \leq \beta \cdot m(\mathcal{O}) \leq \beta \cdot [m^*(E) + \epsilon]. \quad (12)$$

For $1 \leq k \leq n$, we infer from the preceding lemma, applied to the restriction of f to $[c_k, d_k]$, that

$$m^*(E \cap (c_k, d_k)) \leq \frac{1}{\alpha} [f(d_k) - f(c_k)].$$

Therefore, by (11),

$$m^*(E) \leq \sum_{k=1}^n m^*(E \cap (c_k, d_k)) + \epsilon \leq \frac{1}{\alpha} \left[\sum_{k=1}^n [f(d_k) - f(c_k)] \right] + \epsilon. \quad (13)$$

We infer from (12) and (13) that

$$m^*(E) \leq \frac{\beta}{\alpha} \cdot m^*(E) + \frac{1}{\alpha} \cdot \epsilon + \epsilon \text{ for all } \epsilon > 0.$$

Therefore, since $0 \leq m^*(E) < \infty$ and $\beta/\alpha < 1$, $m^*(E) = 0$. □

Lebesgue's Theorem is the best possible in the sense that if E is a set of measure zero contained in the open interval (a, b) , there is an increasing function on (a, b) that fails to be differentiable at each point in E (see Problem 10).

Remark Frigyes Riesz and Béla Sz.-Nagy² remark that Lebesgue's Theorem is "one of the most striking and most important in real variable theory." Indeed, in 1872 Karl Weierstrass presented mathematics with a continuous function on an open interval which failed to be differentiable at any point.³ Further pathology was revealed and there followed a period of uncertainty regarding the spread of pathology in mathematical analysis. Lebesgue's Theorem, which was published in 1904, and its consequences, which we pursue in Section 5, helped restore confidence in the harmony of mathematics analysis.

Let f be integrable over the closed, bounded interval $[a, b]$. Extend f to take the value $f(b)$ on $(b, b+1]$. For $0 < h \leq 1$, define the **divided difference function** $\text{Diff}_h f$ and **average value function** $\text{Av}_h f$ of $[a, b]$ by

$$\text{Diff}_h f(x) = \frac{f(x+h) - f(x)}{h} \text{ and } \text{Av}_h f(x) = \frac{1}{h} \cdot \int_x^{x+h} f \text{ for all } x \in [a, b].$$

By a change of variables in the integral and cancellation, for all $a \leq u < v \leq b$,

$$\int_u^v \text{Diff}_h f = \text{Av}_h f(v) - \text{Av}_h f(u). \quad (14)$$

Corollary 4 Let f be an increasing function on the closed, bounded interval $[a, b]$. Then f' is integrable over $[a, b]$ and

$$\int_a^b f' \leq f(b) - f(a). \quad (15)$$

Proof Since f is increasing on $[a, b+1]$, it is measurable (see Problem 22) and therefore the divided difference functions are also measurable. Lebesgue's Theorem tells us that f is differentiable almost everywhere on (a, b) . Therefore $\{\text{Diff}_{1/n} f\}$ is a sequence of nonnegative measurable functions that converges pointwise almost everywhere on $[a, b]$ to f' . According to Fatou's Lemma,

$$\int_a^b f' \leq \liminf_{n \rightarrow \infty} \left[\int_a^b \text{Diff}_{1/n} f \right] \quad (16)$$

By the change of variable formula (14), for each natural number n , since f is increasing,

$$\int_a^b \text{Diff}_{1/n} f = \frac{1}{1/n} \cdot \int_b^{b+1/n} f - \frac{1}{1/n} \cdot \int_a^{a+1/n} f = f(b) - \frac{1}{1/n} \cdot \int_a^{a+1/n} f \leq f(b) - f(a).$$

Thus

$$\limsup_{n \rightarrow \infty} \left[\int_a^b \text{Diff}_{1/n} f \right] \leq f(b) - f(a). \quad (17)$$

The inequality (15) follows from the inequalities (16) and (17). $\square$

²See page 5 of their book *Functional Analysis* [RSN90].

³A simpler example of such a function, due to Bartel van der Waerden, is examined in Chapter 8 of Patrick Fitzpatrick's *Advanced Calculus* [Fit09].

Remark The integral in (15) is independent of the values taken by f at the endpoints. On the other hand, the right-hand side of this equality holds for the extension of any increasing function of f on the open, bounded interval (a, b) to its closure $[a, b]$. Therefore a tighter form of equality (15) is

$$\int_a^b f' \leq \sup_{x \in (a, b)} f(x) - \inf_{x \in (a, b)} f(x). \quad (18)$$

The right-hand side of this inequality equals $f(b) - f(a)$ if and only if f is continuous at the endpoints. However, even if f is increasing and continuous on $[a, b]$, inequality (15) may be strict. It is strict for the Cantor-Lebesgue function φ on $[0, 1]$ since $\varphi(1) - \varphi(0) = 1$ while φ' vanishes almost everywhere on $(0, 1)$. We show that for an increasing function f on $[a, b]$, (15) is an equality if and only if the function is absolutely continuous on $[a, b]$ (see the forthcoming Corollary 12).

Remark For a continuous function f on a closed, bounded interval $[a, b]$ that is differentiable on the open interval (a, b) , in the absence of a monotonicity assumption on f we cannot infer that its derivative f' is integrable over $[a, b]$. We leave it as an exercise to show that for f defined on $[0, 1]$ by

$$f(x) = \begin{cases} x^2 \sin(1/x^2) & \text{for } 0 < x \leq 1 \\ 0 & \text{for } x = 0, \end{cases}$$

f' is not integrable over $[0, 1]$.

PROBLEMS

5. Show that the Vitali Covering Lemma does not extend to the case in which the covering collection has degenerate closed intervals.
6. Show that the Vitali Covering Lemma does extend to the case in which the covering collection consists of nondegenerate general intervals.
7. Let f be continuous on $\mathbf{R}$. Is there an open interval on which f is monotone?
8. Let I and J be closed, bounded intervals and $\gamma > 0$ be such that $\ell(I) > \gamma \cdot \ell(J)$. Assume $I \cap J \neq \emptyset$. Show that if $\gamma \geq 1/2$, then $J \subseteq 5 * I$, where $5 * I$ denotes the interval with the same center as I and five times its length. Is the same true if $0 < \gamma < 1/2$?
9. Show that a set E of real numbers has measure zero if and only if there is a countable collection of open intervals $\{I_k\}_{k=1}^{\infty}$ for which each point in E belongs to infinitely many of the I_k 's and $\sum_{k=1}^{\infty} \ell(I_k) < \infty$.
10. (Riesz-Nagy) Let E be a set of measure zero contained in the open interval (a, b) . According to the preceding problem, there is a countable collection of open intervals contained in (a, b) , $\{(c_k, d_k)\}_{k=1}^{\infty}$, for which each point in E belongs to infinitely many intervals in the collection and $\sum_{k=1}^{\infty} (d_k - c_k) < \infty$. Define

$$f(x) = \sum_{k=1}^{\infty} \ell((c_k, d_k) \cap (-\infty, x)) \text{ for all } x \text{ in } (a, b).$$

Show that f is increasing and fails to be differentiable at each point in E .

23. Show that a continuous function f on $[a, b]$ is Lipschitz if its upper and lower derivatives are bounded on (a, b) .
24. Show that for f defined in the last remark of this section, f' is not integrable over $[0, 1]$.

6.3 FUNCTIONS OF BOUNDED VARIATION: JORDAN'S THEOREM

Lebesgue's Theorem tells us that a monotone function on an open interval is differentiable almost everywhere. Therefore the *difference* of two increasing functions on an open interval also is differentiable almost everywhere. We now provide a characterization of the class of functions on a closed, bounded interval that may be expressed as the difference of increasing functions, which shows that this class is surprisingly large: it includes, for instance, all Lipschitz functions.

Let f be a real-valued function defined on the closed, bounded interval $[a, b]$ and $P = \{x_0, \dots, x_k\}$ be a partition of $[a, b]$. Define the variation of f with respect to P by

$$V(f, P) = \sum_{i=1}^k |f(x_i) - f(x_{i-1})|,$$

and the **total variation** of f on $[a, b]$ by

$$TV(f) = \sup \{V(f, P) \mid P \text{ a partition of } [a, b]\}.$$

For a subinterval $[c, d]$ of $[a, b]$, $TV(f|_{[c,d]})$ denotes the total variation of the restriction of f to $[c, d]$.

Definition A real-valued function f on the closed, bounded interval $[a, b]$ is said to be of **bounded variation** on $[a, b]$ provided

$$TV(f) < \infty.$$

Example Let f be an increasing function on $[a, b]$. Then f is of bounded variation on $[a, b]$ and

$$TV(f) = f(b) - f(a).$$

Indeed, for any partition $P = \{x_0, \dots, x_k\}$ of $[a, b]$,

$$V(f, P) = \sum_{i=1}^k |f(x_i) - f(x_{i-1})| = \sum_{i=1}^k [f(x_i) - f(x_{i-1})] = f(b) - f(a).$$

Example Let f be a Lipschitz function on $[a, b]$. Then f is of bounded variation of $[a, b]$, and $TV(f) \leq c \cdot (b - a)$, where

$$|f(u) - f(v)| \leq c|u - v| \text{ for all } u, v \text{ in } [a, b].$$

Indeed, for a partition $P = \{x_0, \dots, x_k\}$ of $[a, b]$,

$$V(f, P) = \sum_{i=1}^k |f(x_i) - f(x_{i-1})| \leq c \cdot \sum_{i=1}^k [x_i - x_{i-1}] = c \cdot [b - a].$$

Thus, $c \cdot [b - a]$ is an upper bound of the set of all variations of f with respect to a partition of $[a, b]$ and hence $TV(f) \leq c \cdot [b - a]$.

Example Define the function f on $[0, 1]$ by

$$f(x) = \begin{cases} x \cos(\pi/2x) & \text{if } 0 < x \leq 1 \\ 0 & \text{if } x = 0. \end{cases}$$

Then f is continuous on $[0, 1]$. But f is not of bounded variation on $[0, 1]$. Indeed, for a natural number n , consider the partition $P_n = (0, 1/2n, 1/[2n-1], \dots, 1/3, 1/2, 1)$ of $[0, 1]$. Then

$$V(f, P_n) = 1 + 1/2 + \dots + 1/n.$$

Hence f is not of bounded variation on $[0, 1]$, since the harmonic series diverges.

Observe that if c belongs to (a, b) , P is a partition of $[a, b]$, and P' is the refinement of P obtained by adjoining c to P , then, by the triangle inequality, $V(f, P) \leq V(f, P')$. Thus, in the definition of the total variation of a function on $[a, b]$, the supremum can be taken over partitions of $[a, b]$ that contain the point c . Now a partition P of $[a, b]$ that contains the point c induces, and is induced by, partitions P_1 and P_2 of $[a, c]$ and $[c, b]$, respectively, and for such partitions

$$V(f_{[a,b]}, P) = V(f_{[a,c]}, P_1) + V(f_{[c,b]}, P_2). \quad (19)$$

Take the supremum among such partitions to conclude that

$$TV(f_{[a,b]}) = TV(f_{[a,c]}) + TV(f_{[c,b]}). \quad (20)$$

We infer from this that if f is of bounded variation on $[a, b]$, then

$$TV(f_{[a,v]}) - TV(f_{[a,u]}) = TV(f_{[u,v]}) \geq 0 \text{ for all } a \leq u < v \leq b. \quad (21)$$

Therefore the function $x \mapsto TV(f_{[a,x]})$, which we call the **total variation function** for f , is a real-valued increasing function on $[a, b]$. Moreover, for $a \leq u < v \leq b$, if we take the crudest partition $P = \{u, v\}$ of $[u, v]$, we have

$$f(u) - f(v) \leq |f(v) - f(u)| = V(f_{[u,v]}, P) \leq TV(f_{[u,v]}) = TV(f_{[a,v]}) - TV(f_{[a,u]}).$$

Thus

$$f(v) + TV(f_{[a,v]}) \geq f(u) + TV(f_{[a,u]}) \text{ for all } a \leq u < v \leq b. \quad (22)$$

We have established the following lemma.

Lemma 5 *Let the function f be of bounded variation on the closed, bounded interval $[a, b]$. Then f has the following explicit expression as the difference of two increasing functions on $[a, b]$:*

$$f(x) = [f(x) + TV(f_{[a,x]})] - TV(f_{[a,x]}) \text{ for all } x \in [a, b]. \quad (23)$$

Jordan's Theorem *A function f is of bounded variation on the closed, bounded interval $[a, b]$ if and only if it is the difference of two increasing functions on $[a, b]$.*

Proof Let f be of bounded variation on $[a, b]$. The preceding lemma provides an explicit representation of f as the difference of increasing functions. To prove the converse, let $f = g - h$ on $[a, b]$, where g and h are increasing functions on $[a, b]$. For any partition $P = \{x_0, \dots, x_k\}$ of $[a, b]$,

$$\begin{aligned} V(f, P) &= \sum_{i=1}^k |f(x_i) - f(x_{i-1})| \\ &= \sum_{i=1}^k |[g(x_i) - g(x_{i-1})] + [h(x_{i-1}) - h(x_i)]| \\ &\leq \sum_{i=1}^k |g(x_i) - g(x_{i-1})| + \sum_{i=1}^k |h(x_{i-1}) - h(x_i)| \\ &= \sum_{i=1}^k [g(x_i) - g(x_{i-1})] + \sum_{i=1}^k [h(x_i) - h(x_{i-1})] \\ &= [g(b) - g(a)] + [h(b) - h(a)]. \end{aligned}$$

Thus, the set of variations of f with respect to partitions of $[a, b]$ is bounded above by $[g(b) - g(a)] + [h(b) - h(a)]$ and therefore f is of bounded variation of $[a, b]$. $\square$

We call the expression of a function of bounded variation f as the difference of increasing functions a **Jordan decomposition** of f .

Corollary 6 If the function f is of bounded variation on the closed, bounded interval $[a, b]$, then it is differentiable almost everywhere on the open interval (a, b) and f' is integrable over $[a, b]$.

Proof According to Jordan's Theorem, f is the difference of two increasing functions on $[a, b]$. Thus Lebesgue's Theorem tells us that f is the difference of two functions which are differentiable almost everywhere on (a, b) . Therefore f is differentiable almost everywhere on (a, b) . The integrability of f' follows from Corollary 4. $\square$

PROBLEMS

25. Suppose f is continuous on $[0, 1]$. Must there be a nondegenerate closed subinterval $[a, b]$ of $[0, 1]$ for which the restriction of f to $[a, b]$ is of bounded variation?
26. Let f be the Dirichlet function, the characteristic function of the rationals in $[0, 1]$. Is f of bounded variation on $[0, 1]$?
27. Define $f(x) = \sin x$ on $[0, 2\pi]$. Find two increasing functions h and g for which $f = h - g$ on $[0, 2\pi]$.
28. Let f be a step function on $[a, b]$. Find a formula for its total variation.
29. (a) Define

$$f(x) = \begin{cases} x^2 \cos(1/x^2) & \text{if } x \neq 0, x \in [-1, 1] \\ 0 & \text{if } x = 0. \end{cases}$$

Is f of bounded variation on $[-1, 1]$?

(b) Define

$$g(x) = \begin{cases} x^2 \cos(1/x) & \text{if } x \neq 0, x \in [-1, 1] \\ 0 & \text{if } x = 0. \end{cases}$$

Is g of bounded variation on $[-1, 1]$?

30. Show that the linear combination of two functions of bounded variation is also of bounded variation. Is the product of two such functions also of bounded variation?
31. Let P be a partition of $[a, b]$ that is a refinement of the partition P' . For a real-valued function f on $[a, b]$, show that $V(f, P') \leq V(f, P)$.
32. Assume f is of bounded variation on $[a, b]$. Show that there is a sequence of partitions $\{P_n\}$ of $[a, b]$ for which the sequence $\{TV(f, P_n)\}$ is increasing and converges to $TV(f)$.
33. Let $\{f_n\}$ be a sequence of real-valued functions on $[a, b]$ that converges pointwise on $[a, b]$ to the real-valued function f . Show that

$$TV(f) \leq \liminf TV(f_n).$$

34. Let f and g be of bounded variation on $[a, b]$. Show that

$$TV(f + g) \leq TV(f) + TV(g) \text{ and } TV(\alpha f) = |\alpha|TV(f).$$

35. For α and β positive numbers, define the function f on $[0, 1]$ by

$$f(x) = \begin{cases} x^\alpha \sin(1/x^\beta) & \text{for } 0 < x \leq 1 \\ 0 & \text{for } x = 0. \end{cases}$$

Show that if $\alpha > \beta$, then f is of bounded variation on $[0, 1]$, by showing that f' is integrable over $[0, 1]$. Then show that if $\alpha \leq \beta$, then f is not of bounded variation on $[0, 1]$.

36. Let f fail to be of bounded variation on $[0, 1]$. Show that there is a point x_0 in $[0, 1]$ such that f fails to be of bounded variation on each nondegenerate closed subinterval of $[0, 1]$ that contains x_0 .

6.4 ABSOLUTELY CONTINUOUS FUNCTIONS

Definition A real-valued function f on a closed, bounded interval $[a, b]$ is said to be **absolutely continuous** on $[a, b]$ provided for each $\epsilon > 0$, there is a $\delta > 0$ such that for every finite disjoint collection $\{(a_k, b_k)\}_{k=1}^n$ of open intervals in (a, b) ,

$$\text{if } \sum_{k=1}^n [b_k - a_k] < \delta, \text{ then } \sum_{k=1}^n |f(b_k) - f(a_k)| < \epsilon.$$

The criterion for absolute continuity in the case the finite collection of intervals consists of a single interval is the criterion for the uniform continuity of f on $[a, b]$. Thus absolutely continuous functions are continuous. The converse is false, even for increasing functions.

Example The Cantor-Lebesgue function φ is increasing and continuous on $[0, 1]$, but it is not absolutely continuous (see also Problems 40 and 48). Indeed, to see that φ is not absolutely continuous, let n be a natural number. At the n -th stage of the construction of the Cantor set, a disjoint collection $\{[c_k, d_k]\}_{1 \leq k \leq 2^n}$ of 2^n subintervals of $[0, 1]$ have been constructed that cover the Cantor set, each of which has length $(1/3)^n$. The Cantor-Lebesgue function is constant on each of the intervals that comprise the complement in $[0, 1]$ of this collection of intervals. Therefore, since φ is increasing and $\varphi(1) - \varphi(0) = 1$,

$$\sum_{1 \leq k \leq 2^n} [d_k - c_k] = (2/3)^n \text{ while } \sum_{1 \leq k \leq 2^n} [\varphi(d_k) - \varphi(c_k)] = 1.$$

There is no response to the $\epsilon = 1$ challenge regarding the criterion for φ to be absolutely continuous.

Clearly linear combinations of absolutely continuous functions are absolutely continuous. However, the composition of absolutely continuous functions may fail to be absolutely continuous (see Problems 43, 44, and 45).

Proposition 7 *If the function f is Lipschitz on a closed, bounded interval $[a, b]$, then it is absolutely continuous on $[a, b]$.*

Proof Let $c > 0$ be a Lipschitz constant for f on $[a, b]$, that is,

$$|f(u) - f(v)| \leq c|u - v| \text{ for all } u, v \in [a, b].$$

Then, regarding the criterion for the absolute continuity of f , it is clear that $\delta = \epsilon/c$ responds to any $\epsilon > 0$ challenge. $\square$

There are absolutely continuous functions that fail to be Lipschitz: the function f on $[0, 1]$, defined by $f(x) = \sqrt{x}$ for $0 \leq x \leq 1$, is absolutely continuous but not Lipschitz (see Problem 37).

Theorem 8 *Let the function f be absolutely continuous on the closed, bounded interval $[a, b]$. Then f is the difference of increasing absolutely continuous functions and, in particular, is of bounded variation.*

Proof We first prove that f is of bounded variation. Indeed, let δ respond to the $\epsilon = 1$ challenge regarding the criterion for the absolute continuity of f . Let P be a partition of $[a, b]$ into N closed intervals $\{[c_k, d_k]\}_{k=1}^N$, each of length less than δ . Then, by the definition of δ in relation to the absolute continuity of f , it is clear that $TV(f|_{[c_k, d_k]}) \leq 1$, for $1 \leq k \leq N$. The additivity formula (19) extends to finite sums. Hence

$$TV(f) = \sum_{k=1}^N TV(f|_{[c_k, d_k]}) \leq N.$$

Therefore f is of bounded variation. In view of (23) and the absolute continuity of sums of absolutely continuous functions, to show that f is the difference of increasing absolutely continuous functions it suffices to show that the total variation function for f is absolutely

continuous. Let $\epsilon > 0$. Choose δ as a response to the $\epsilon/2$ challenge regarding the criterion for the absolute continuity of f on $[a, b]$. Let $\{(c_k, d_k)\}_{k=1}^n$ be a disjoint collection of open subintervals of (a, b) for which $\sum_{k=1}^n [d_k - c_k] < \delta$. For $1 \leq k \leq n$, let P_k be a partition of $[c_k, d_k]$. By the choice of δ in relation to the absolute continuity of f on $[a, b]$,

$$\sum_{k=1}^n TV(f|_{[c_k, d_k]}, P_k) < \epsilon/2.$$

Take the supremum as, for $1 \leq k \leq n$, P_k vary among partitions of $[c_k, d_k]$, to obtain

$$\sum_{k=1}^n TV(f|_{[c_k, d_k]}) \leq \epsilon/2 < \epsilon.$$

We infer from (21) that, for $1 \leq k \leq n$, $TV(f|_{[c_k, d_k]}) = TV(f|_{[a, d_k]}) - TV(f|_{[a, c_k]})$. Hence

$$\text{if } \sum_{k=1}^n [d_k - c_k] < \delta, \text{ then } \sum_{k=1}^n |TV(f|_{[a, d_k]}) - TV(f|_{[a, c_k]})| < \epsilon. \quad (24)$$

Therefore the total variation function for f is absolutely continuous on $[a, b]$. $\square$

Theorem 9 Let the function f be continuous on the closed, bounded interval $[a, b]$. Then f is absolutely continuous on $[a, b]$ if and only if the family of divided difference functions $\{\text{Diff}_h f\}_{0 < h \leq 1}$ is uniformly integrable over $[a, b]$.

Proof. First assume $\{\text{Diff}_h f\}_{0 < h \leq 1}$ is uniformly integrable over $[a, b]$. Let $\epsilon > 0$. Choose $\delta > 0$ for which

$$\int_E |\text{Diff}_h f| < \epsilon/2 \text{ if } m(E) < \delta \text{ and } 0 < h \leq 1.$$

We claim that δ responds to the ϵ challenge regarding the criterion for f to be absolutely continuous. Indeed, let $\{(c_k, d_k)\}_{k=1}^n$ be a disjoint collection of open subintervals of (a, b) for which $\sum_{k=1}^n [d_k - c_k] < \delta$. For $0 < h \leq 1$ and $1 \leq k \leq n$, by (14),

$$Av_h f(d_k) - Av_h f(c_k) = \int_{c_k}^{d_k} \text{Diff}_h f.$$

Therefore

$$\sum_{k=1}^n |Av_h f(d_k) - Av_h f(c_k)| \leq \sum_{k=1}^n \int_{c_k}^{d_k} |\text{Diff}_h f| = \int_E |\text{Diff}_h f|,$$

where $E = \bigcup_{k=1}^n (c_k, d_k)$ has measure less than δ . Thus, by the choice of δ ,

$$\sum_{k=1}^n |Av_h f(d_k) - Av_h f(c_k)| < \epsilon/2 \text{ for all } 0 < h \leq 1.$$

Since f is continuous, take the limit as $h \rightarrow 0^+$ to obtain

$$\sum_{k=1}^n |f(d_k) - f(c_k)| \leq \epsilon/2 < \epsilon.$$

Hence f is absolutely continuous.

To prove the converse, suppose f is absolutely continuous. The preceding theorem tells us that f is the difference of increasing absolutely continuous functions. We may therefore assume that f is increasing, so that the divided difference functions are nonnegative. To verify the uniform integrability of $(\text{Diff}_h f)_{0 < h \leq 1}$, let $\epsilon > 0$. We must show that there is a $\delta > 0$ such that for each measurable subset E of (a, b) ,

$$\int_E \text{Diff}_h f < \epsilon \text{ if } m(E) < \delta \text{ and } 0 < h \leq 1. \quad (25)$$

According to Theorem 11 of Chapter 2, a measurable set E is contained in a G_δ set G for which $m(G \setminus E) = 0$. But every G_δ set is the intersection of a descending sequence of open sets. Moreover, every open set is the disjoint union of a countable collection of open intervals, and therefore every open set is the union of an ascending sequence of open sets, each of which is the union of a finite disjoint collection of open intervals. Therefore, by the continuity of integration, to verify (25) it suffices to find a $\delta > 0$ such that for $\{(c_k, d_k)\}_{k=1}^n$ a disjoint collection of open subintervals of (a, b) ,

$$\int_E \text{Diff}_h f < \epsilon/2 \text{ if } m(E) < \delta, \text{ where } E = \bigcup_{k=1}^n (c_k, d_k), \text{ and } 0 < h \leq 1. \quad (26)$$

Choose $\delta > 0$ as the response to the $\epsilon/2$ challenge regarding the criterion for the absolute continuity of f on $[a, b + 1]$. By a change of variables for the Riemann integral and cancellation,

$$\int_u^v \text{Diff}_h f = \frac{1}{h} \int_0^h g(t) dt, \text{ where } g(t) = f(v+t) - f(u+t) \text{ for } 0 \leq t \leq 1 \text{ and } a \leq u < v \leq b.$$

Therefore, if $\{(c_k, d_k)\}_{k=1}^n$ is a disjoint collection of open subintervals of (a, b) ,

$$\int_E \text{Diff}_h f = \frac{1}{h} \cdot \int_0^h g(t) dt,$$

where

$$E = \bigcup_{k=1}^n (c_k, d_k) \text{ and } g(t) = \sum_{k=1}^n [f(d_k+t) - f(c_k+t)] \text{ for all } 0 \leq t \leq 1.$$

If $\sum_{k=1}^n [d_k - c_k] < \delta$, then, for $0 \leq t \leq 1$, $\sum_{k=1}^n [(d_k+t) - (c_k+t)] < \delta$, and therefore $g(t) < \epsilon/2$.
Thus

$$\int_E \text{Diff}_h f = \frac{1}{h} \cdot \int_0^h g(t) dt < \epsilon/2.$$

Hence (26) is verified for this choice of δ . $\square$

Remark For a nondegenerate closed, bounded interval $[a, b]$, let $\mathcal{F}_{Lip}$, $\mathcal{F}_{AC}$, and $\mathcal{F}_{BV}$ denote the families of functions on $[a, b]$ that are Lipschitz, absolutely continuous, and of bounded variation, respectively. We have the following strict inclusions:

$$\mathcal{F}_{Lip} \subsetneq \mathcal{F}_{AC} \subsetneq \mathcal{F}_{BV}. \quad (27)$$